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# Hyunsu Lee

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## EDUCATION

### Doctorate of Medicine(Physiology)

03 2007 - 02 2013

*Seoul National University College of Medicine*

Thesis: Role of GABA-induced  $\text{Ca}^{2+}$  signal in axonal outgrowth in hippocampal newborn granule cells

### Bachelor of Medicine

03 2001 - 02 2007

*Seoul National University College of Medicine*

Physician Certificate ID: 92269

## WORK AND RESEARCH EXPERIENCE

### Associate Professor

09 2026 - Present

*Department of Physiology and BioMedical Informatics, Pusan National University School of Medicine  
Yangsan, Republic of Korea*

### Assistant Professor

09 2023 - 08 2026

*Department of Physiology, Pusan National University School of Medicine*

*Yangsan, Republic of Korea*

### Assistant Professor

03 2016 - 08 2023

*Department of Medical Informatics (03 2022 - 08 2023), Department of Anatomy (03 2016 - 02 2022),  
Keimyung University School of Medicine*

*Daegu, Republic of Korea*

### Teaching Assistant and Research Fellow

03 2014 - 02 2016

*Keimyung University School of Medicine*

*Daegu, Republic of Korea*

### Intern Doctor

03 2013 - 07 2013

*Seoul National University Hospital*

*Seoul, Republic of Korea*

### Research Doctor

03 2011 - 02 2013

*Department of Physiology, Seoul National University College of Medicine*

*Seoul, Republic of Korea*

### Teaching Assistant

09 2007 - 08 2009

*Department of Physiology, Seoul National University College of Medicine*

*Seoul, Republic of Korea*

## PUBLICATIONS(SELECTED)

- Lee, H. (2024). The rise of ChatGPT: Exploring its potential in medical education. *Anat. Sci. Educ.* 17(5), 926-931. doi:10.1002/ase.2270
- Lee, H. (2022). Toward the biological model of the hippocampus as the successor representation agent. *Biosystems*, 104612. doi:10.1016/j.biosystems.2022.104612
- Seo, I., Park, S.-H., & Lee, H. (2026). Eligibility-trace-gated depression in predecessor feature learning enables post-reward exploration. *Cognitive Neurodynamics*, 20(1), 0. <https://doi.org/10.1007/s11571-026-10477-5>
- Avila, F., Kim, J., Park, J.-C., Kang, R., Lee, H., & Park, S.-H. (2026). CASCADE: criticality avalanche spike cross-platform analysis detection engine, a multi-manufacturer MEA bash analysis pipeline. *Bioinformatics*, 42(8), btg570. <https://doi.org/10.1093/bioinformatics/btg570> (Lee, H. and Park, S.-H. are corresponding authors)
- Cho, E., Kwon, J., Lee, G., Shin, J., Lee, H., Lee, S.-H., Chung, C. K., Yoon, J., & Ho, W.-K. (2024). Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy. *Nature Communications*, 15(1), 6683. doi.org/10.1038/s41467-024-51065-7

## RESEARCH INTEREST

- Learning and memory mechanism on neural microcircuit level
- Spatial navigation of hippocampus
- Convergence of neuroscience and AI
- Application of AI on biomedical area and its ethical issue

## HONOR AWARDS

Young Faculty Teaching Award (2026)

Basic Researcher Award (2015)

Poster Award (2014)

Poster Award (2010) *Synapse Division of Korean Society Biochemistry and Molecular Biology*

Research Award (2009)

*Pusan National University School of Medicine*

*Korean Medical Association*

*Korean Association of Anatomist*

*Association of Korean Neuroscientists*

## SKILLS

**Programming** Python, Git, L<sup>A</sup>T<sub>E</sub>X, Unix CLT, Markdown, R

**Scientific** Electrophysiology, Patch-clamp, Molecular biology etc

**Communication** Korean (native), English (business)

## PUBLICATIONS

63. Avila, F., Kim, J., Park, J.-C., Kang, R., **Lee, H.**, & Park, S.-H. (2026). CASCADE: criticality avalanche spike cross-platform analysis detection engine, a multi-manufacturer MEA bash analysis pipeline. *Bioinformatics*, 42(8), btac570. <https://doi.org/10.1093/bioinformatics/btac570> (**Lee, H.** and Park, S.-H. are corresponding authors)
62. Oh, S., Han, I. H., Lee, J. I., Choi, B. K., & **Lee, H.** (2026). Diagnosis classification in EMR data using latent representations and SNOMED-CT mapping for improved medical data integration. *Medical & Biological Engineering & Computing*, 0. <https://doi.org/10.1007/s11517-026-03654-5>
61. Jeon, J., Jang, J., Moon, C. Y., Lee, J., Hong, V. S., Kang, H., Park, J. Y., Heo, N. H., Park, J.-W., Park, J.-H., Lee, J.-H., Lee, H. W., Bae, S. U., **Lee, H.**, & Kim, S. (2026). Evaluation of the Anti-Cancer Effects of KMU-11342 in In Vitro and Ex Vivo Models of Colorectal Cancer. *Pharmaceuticals (Basel)*, 19(7), 985. <https://doi.org/10.3390/ph19070985> (**Lee, H.** and Kim, S. are corresponding authors)
60. Seo, I., Park, S.-H., & **Lee, H.** (2026). Eligibility-trace-gated depression in predecessor feature learning enables post-reward exploration. *Cognitive Neurodynamics*, 20(1), 0. <https://doi.org/10.1007/s11571-026-10477-5>
59. Oh, S., Jang, J., Ko, Y., **Lee, H.**, & Lim, S. (2026). Intuitive Risk Equation for Post-Transplant Bloodstream Infection Prediction: A Symbolic Regression Approach. *Biomedicines*, 14(4), 840. <https://doi.org/10.3390/biomedicines14040840> (**Lee, H.** and Lim, S. are corresponding authors)
58. Kim, D.-E., Nguyen, H.-T., Mai, N.-D., Bao, L., & **Lee, H.** (2026). End-to-End EEG-Based Stress Detection: A Benchmark Using Repurposed DEAP and Russell's Circumplex Model. *IEEE Access*, 14, 24127–24138. <https://doi.org/10.1109/access.2026.3662465>
57. Baek, H. S., Yoon, D., Hong, V. S., Lee, S.-J., Lee, S.-I., Lee, J.-Y., Lee, H. W., Jeong, S., Jung, H., Kang, H., Kim, S. H., Son, C.-N., **Lee, H.**, Lee, J., & Kim, S. (2026). A novel small molecule KMU-11361 attenuates rheumatoid arthritis by mechanistic inhibition of the TAK1–NF- $\kappa$ B–NLRP3 axis. *Inflammation Research*, 75(1), 0. <https://doi.org/10.1007/s00011-026-02197-7> (**Lee, H.**, Lee, J. and Kim, S. are corresponding authors)
56. Montecillo, J. A. V., Yoo, H. J., Lee, Y.-Y., Park, C. M., Cho, A., **Lee, H.**, Yoon, H. Y., Kim, J. M., Lee, N. Y., Park, S.-H., Park, N. J.-Y., Han, H. S., Seo, I., & Chong, G. O. (2025). Pangenome analysis of *Lactobacillus mulieris* strains reveals distinct subspecies clusters with defined ecological adaptations. *Microbiol Spectr*, 13(11), e0201125. <https://doi.org/10.1128/spectrum.02011-25>
55. **Lee, H.**, & Daou, A. (2025). Hippocampal Function and Reinforcement Learning. *Frontiers in Computational Neuroscience*, 19, 1595369. <https://doi.org/10.3389/fncom.2025.1595369>

54. Montecillo, J. A. V., Lee, J. H., Yoo, H. J., Lee, Y.-Y., Park, C. M., Cho, A., **Lee, H.**, Kim, J. M., Lee, N. Y., Park, S.-H., Park, N. J.-Y., Han, H. S., Chong, G. O., & Seo, I. (2025). Characterization of *Streptococcus mitis* subsp. *carlssonii* isolated from human vagina: prevalence, phenotypic, and genomic insights. *Front Microbiol*, 16, 1625724. <https://doi.org/10.3389/fmicb.2025.1625724>
53. Yoon, D., & **Lee, H.** (2025). In silico discovery of novel compounds for FAK activation using virtual screening, AI-based prediction, and molecular dynamics. *Computational Biology and Chemistry*, 108420. <https://doi.org/10.1016/j.compbiolchem.2025.108420>
52. **Lee, H.** (2025). Noise Resilience of Successor and Predecessor Feature Algorithms in One- and Two-Dimensional Environments. *Sensors*, 25(9), 979. <https://doi.org/10.3390/s25030979>
51. **Lee, H.** (2024). The rise of ChatGPT: Exploring its potential in medical education. *Anat. Sci. Educ.* 17(5), 926-931. [doi.org/10.1002/ase.2270](https://doi.org/10.1002/ase.2270)
50. Heo, Y.-R., **Lee, H.**, Lee, S.-W., Kim, B.-S., Kim, H.-T., & Lee, J.-H. (2024). Supernumerary Head of the Biceps Brachii Muscle Influences the Topography of the Coracobrachialis and Biceps Brachii Muscles. *Medicina*, 60(11), 1726. [doi.org/10.3390/medicina60111726](https://doi.org/10.3390/medicina60111726)
49. Lim, J. S., Kim, S., Park, M.-N., **Lee, H.**, Baek, H. S., Kim, J. K., Kim, H. W., & Hong, J.-H. (2024). The Neuroprotective Effect of Therapeutic Hypothermia in Cognitive Impairment of an Ischemia/Reperfusion Injury Mouse Model. *Medicina*, 60(3), 350. [doi.org/10.3390/medicina60030350](https://doi.org/10.3390/medicina60030350)
48. Cho, E., Kwon, J., Lee, G., Shin, J., **Lee, H.**, Lee, S.-H., Chung, C. K., Yoon, J., & Ho, W.-K. (2024). Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy. *Nature Communications*, 15(1), 6683. [doi.org/10.1038/s41467-024-51065-7](https://doi.org/10.1038/s41467-024-51065-7)
47. **Lee, H.** (2024). Reflections on the Development of Large Language Models and Their Impact on Basic Medical Education. *Anat. Biol. Anthropol.*, 37(2), 59-67. [doi.org/10.11637/aba.2024.37.2.59](https://doi.org/10.11637/aba.2024.37.2.59)
46. Montecillo, J. A. V., Yoo, H. J., Lee, Y.-Y., Park, C., Cho, A., **Lee, H.**, Park, N. J.-Y., Han, H. S., Chong, G. O., & Seo, I. (2024). Cell-Free Supernatant of Vaginal Viridans *Streptococcus* Induces Membrane Permeabilization and Transcriptional Regulation in Methicillin-Resistant *Staphylococcus aureus*. *Probiotics and Antimicrobial Proteins*. [doi.org/10.1007/s12602-024-10412-2](https://doi.org/10.1007/s12602-024-10412-2)
45. **Lee, H.** (2023). Tuning the Weights: The Impact of Initial Matrix Configurations on Successor Features' Learning Efficacy. *Electronics*, 12(20), 4212.
44. Seo, I., & **Lee, H.** (2024). Investigating Transfer Learning in Noisy Environments: A Study of Predecessor and Successor Features in Spatial Learning Using a T-Maze. *Sensors*, 24(19), 6419. [doi.org/10.3390/s24196419](https://doi.org/10.3390/s24196419)
43. Kang, S.-J., Kim, Y.-H., Nguyen-Phuong, T., Kim, Y., Oh, J.-M., Go, J.-c., Kim, D., Park, C.-G., **Lee, H.**, & Kim, H. J. (2024). Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8<sup>+</sup> T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease. *Journal of Neuroimmunology*, 578488. [doi.org/10.1016/j.jneuroim.2024.578488](https://doi.org/10.1016/j.jneuroim.2024.578488) (Park, C.-G., **Lee, H.**, and Kim, H. J. are corresponding authors)
42. **Lee, H.**, Park, M.-N., Lim, J. S., Baek, H. S., Kim, S., Kim, H. W., Hong, J.-H. (2023). The Effect of Hypothermia on Cognitive Impairment and Anxiety-like Behavior in a Cardiac Arrest Mouse Model. *Quantitative Bio-Science*, 42(1), 9-16. [doi.org/10.22283/qbs.2023.42.1.9](https://doi.org/10.22283/qbs.2023.42.1.9)
41. Shin, K., Lee, J. S., Lee, J. Y., **Lee, H.**, Kim, J., Byeon, J. S., Jung, H. Y., Kim, D. H., & Kim, N. (2023). An Image Turing Test on Realistic Gastroscopy Images Generated by Using the Progressive Growing of Generative Adversarial Networks. *J Digit Imaging*. [doi:10.1007/s10278-023-00803-2](https://doi.org/10.1007/s10278-023-00803-2)
40. Lim, D. M., **Lee, H.**, Eom, K., Kim, Y. H., Kim, S. (2023). Bioinformatic analysis of the obesity paradox and possible associated factors in colorectal cancer using TCGA cohorts. *Journal of Cancer*, 14(3), 379-392. (Lim, D. M. and Lee, H. has equally contributed)

39. Yim, H., **Lee, H.** (2022). A Critical Review on the Argument that the Sacral Division of the Autonomic Nervous System should be Classified as the Sympathetic Nervous System. *Anatomy & Biological Anthropology*, 35(4), 135-143. doi:doi.org/10.11637/aba.2022.35.4.135
38. **Lee, H.** (2022). Toward the biological model of the hippocampus as the successor representation agent. *Biosystems*, 104612. doi:10.1016/j.biosystems.2022.104612
37. Bae, A.-N., Jung, S.-J., Lee, J.-H., **Lee, H.**, Park, S. G. (2022). Clinical Value of EZH2 in Hepatocellular Carcinoma and Its Potential for Target Therapy. *Medicina*, 58(2). doi:10.3390/medicina58020155
36. Park, K. R., Park, W.-J., Lee, S.-W., Kim, H., **Lee, H.**, Lee, J.-H. (2022). A Simple Method to Locate the Master Knot of Henry Using the Correlation between the Flexor Tendon Length Parameter and the Foot Length. *International Journal of Environmental Research and Public Health*, 19(4), 2281.
35. Yang, K., **Lee, H.**, Choi, I.-J., Jeong, W., Kim, H.-T., Wei, Q., Lee, J.-H. (2021). Topography and Anatomical Variations of the Axillary Artery. *BioMed Research International*, 2021, 1-8. doi:10.1155/2021/6393780
34. Lee, J. S., Yun, J., Ham, S., Park, H., **Lee, H.**, Kim, J., Kim, D. H. (2021). Machine learning approach for differentiating cytomegalovirus esophagitis from herpes simplex virus esophagitis. *Sci Rep*, 11(1), 3672. doi:10.1038/s41598-020-78556-z
33. **Lee, H.**, Kim, Y.-T. (2020). Depression and Anxiety Disorder in Perspective of Neurogenesis and Pattern Separation. *J Korean Soc Biol Ther Psychiatry*, 26(1), 5-13.
32. Seo, I., **Lee, H.** (2019). Predicting transgenic markers of a neuron by electrophysiological properties using machine learning. *Brain Research Bulletin*, 150, 102-110. doi:10.1016/j.brainresbull.2019.05.012
31. Yang, K., Park, S.-U., Kim, J.-Y., **Lee, H.**, Choi, I.-J., Lee, J.-H. (2019). Effectiveness of Smartphone Application in Histology Practice. *Anatomy & Biological Anthropology*, 32(1), 17. doi:10.11637/aba.2019.32.1.17
30. **Lee, H.**, Ahn, D., Sohn, J. H., Kim, Y.-H., Lee, J.-H., Kim, H. (2019). Different correlations between tumor size and cancer-related gene profiles according to histologic type of salivary gland tumor. *Int J Clin Exp Pathol*, 12(7), 2809-2816.
29. Kim, H. J., Seo, I., Shim, W., Lee, C.-E., Chang, W., **Lee, H.**, Park, S.-H. (2018). Three Dimensional Nucleus Imaging analysis & in vitro Genotoxicity Test through Advanced Cytokinesis-Block Micronucleus Assay. *Journal of Alternatives to Animal Experiments*, 12(1), 33-39.
28. Kim, H., Ahn, D., Sohn, J. H., Kim, Y.-H., Lee, J.-H., **Lee, H.** (2018). TERT Promoter Mutation and Telomere Length in Salivary Gland Tumors. *Pathology oncology research : POR*, 24(3), 697-698. doi:10.1007/s12253-017-0275-6
27. **Lee, H.**, Cho, J.-H., Park, W.-J., Jung, S.-J., Choi, I.-J., Lee, J.-H. (2018). Loss of the Association between Telomere Length and Mitochondrial DNA Copy Number Contribute to Colorectal Carcinogenesis. *Pathology oncology research : POR*, 24(2), 323-328. doi:10.1007/s12253-017-0245-z
26. Yang, K., Park, J.-H., Jung, S.-J., **Lee, H.**, Choi, I.-J., Lee, J.-H. (2017). Topography of the Middle Genicular Artery is Associated with the Superior and Inferior Genicular Arteries. *International Journal of Morphology*, 35(3), 913-918. doi:10.4067/s0717-95022017000300019
25. Oh, J. W., **Lee, H.**, Lee, J.-H. (2017). Topography of the coracobrachialis muscle and the musculocutaneous nerve by muscular variation. *Surgical and radiologic anatomy : SRA*, 39(1), 115-116. doi:10.1007/s00276-016-1721-y
24. Yang, K., Jung, S. J., **Lee, H.**, Choi, I. J., Lee, J. H. (2017). Topographical relations between the Gantzer's muscle and neurovascular structures. *Surg Radiol Anat*, 39(8), 843-848. doi:10.1007/s00276-016-1803-x

23. Park, J.-H., Lee, M.-Y., Yang, K., Lee, J.-H., Choi, I.-J., **Lee, H.** (2017). The Application and Effectiveness for Medical Procedural Skills through the Use of the Cadaver Model. *Korean Journal of Physical Anthropology*, 30(1), 21-27. doi:10.11637/kjpa.2017.30.1.21
22. Jung, S.-J., Kim, D.-S., Park, W.-J., **Lee, H.**, Choi, I.-J., Park, J.-Y., Lee, J.-H. (2017). Mutation of the TERT promoter leads to poor prognosis of patients with non-small cell lung cancer. *Oncology Letters*, 14(2), 1609-1614.
21. Kim, H., Ahn, D., Sohn, J. H., Kim, Y.-H., Lee, J.-H., **Lee, H.** (2017). Erratum to: TERT Promoter Mutation and Telomere Length in Salivary Gland Tumors. *Pathology oncology research : POR*, 1. doi:10.1007/s12253-017-0294-3
20. Lee, S.-W., Lee, J.-H., **Lee, H.** (2017). Double Gantzer's Muscles by Four Muscle Bellies and Its Clinical Significance: A Case Report., 1-4. doi:10.11637/kjpa.2017.30.2.67
19. **Lee, H.** (2017). Comparison of the Electrophysiological Properties of Excitatory and Inhibitory Neurons in the Mouse Visual Cortex. *Quantitative Bio-Science*, 36(1), 65-69. doi:10.22283/qbs.2017.36.1.65
18. Wang, Q., Oh, J. W., Lee, H.-L., Dhar, A., Peng, T., Ramos, R., Guerrero-Juarez, C. F., Wang, X., Zhao, R., Cao, X., Le, J., Fuentes, M. A., Jocoy, S. C., Rossi, A. R., Vu, B., Pham, K., Wang, X., Mali, N. M., Park, J. M., , **Lee, H.**, ,Plikus, M. V. (2017). A multi-scale model for hair follicles reveals heterogeneous domains driving rapid spatiotemporal hair growth patterning.. *eLife*, 6, 707. doi.org/10.7554/eLife.22772
17. **Lee, H.**, Jin, J.-D., La, B.-M., Park, W.-J., Choi, I.-J., Lee, J.-H. (2016). TERT promoter mutation, telomere length, and TERT expression in gastric cancer. *Int J Clin Exp Pathol*, 9(2), 1758-1763.
16. Jung, S.-J., **Lee, H.**, Choi, I.-J., Lee, J.-H. (2016). Muscular axillary arch accompanying variation of the musculocutaneous nerve: axillary arch. *Anatomy & cell biology*, 49(2), 160. doi:10.5115/acb.2016.49.2.160
15. **Lee, H.**, Moon, Y.-S., Park, H. S., Kim, H.-T., Choi, I.-J. (2016). A radial artery originating from the thoracoacromial artery. *Surgical and radiologic anatomy : SRA*, 38(5), 631-633. doi:10.1007/s00276-015-1576-7
14. **Lee, H.**, Lee, S.-H., Kim, S.-J., Choi, W.-I., Lee, J.-H., Choi, I.-J. (2015). Variations of the cubital superficial vein investigated by using the intravenous illuminator. *Anatomy & cell biology*, 48(1), 62-65. doi:10.5115/acb.2015.48.1.62
13. Yang, K., **Lee, H.**, Lee, J.-H., Choi, I.-J. (2015). Reply: Anomalous Muscles in Carpal Tunnel Associated with Neurovascular Variation: Case Report and Brief Review. *Forensic Medicine and Anatomy Research*, 03(01), 25-26. doi:10.4236/fmar.2015.31005
12. **Lee, H.**, Lee, J.-H., Kim, D.-K., Choi, I.-J., Hwang, I., Kang, Y.-N., Kim, S. (2015). PIK3CA Amplification Is Common in Left Side-Tubular Adenomas but Uncommon Sessile Serrated Adenomas Exclusively with KRAS Mutation. *International Journal of Medical Sciences*, 12(4), 349-353. doi:10.7150/ijms.11281
11. **Lee, H.**, Kim, H.-T., Lee, J.-H., Choi, I.-J. (2015). Low frequency of the lateral thoracic artery originating from the thoracoacromial artery. *Surgical and radiologic anatomy : SRA*, 37(3), 319-320. doi:10.1007/s00276-014-1363-x
10. **Lee, H.**, Lee, J.-H., Kim, D.-C., Hwang, I., Kang, Y.-N., Gwon, G.-J., Choi, I.-J., Kim, S. (2015). Is mitochondrial DNA copy number associated with clinical characteristics and prognosis in gastric cancer. *Asian Pacific journal of cancer prevention : APJCP*, 16(1), 87-90.
9. Lee, K.-H., **Lee, H.**, Yang, C. H., Ko, J.-S., Park, C.-H., Woo, R.-S., Kim, J. Y., Sun, W., Kim, J.-H., Ho, W.-K., Lee, S.-H. (2015). Bidirectional Signaling of Neuregulin-2 Mediates Formation of GABAergic Synapses and Maturation of Glutamatergic Synapses in Newborn Granule Cells of Postnatal Hippocampus. *The Journal of Neuroscience*, 35(50), 16479-16493.

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7. **Lee, H.**, La, B.-M., Hwang, I., Kang, Y.-N., Choi, I.-J., Lee, J.-H. (2015). Absence of GNAS mutation in colorectal carcinogenesis. *Tumori*, 0. doi:10.5301/tj.5000400
6. **Lee, H.**, Yang, K., Choi, I.-J., Lee, J.-H. (2014). The ileocolic vein passing through the ring-shaped part of ileocolic artery. *Surgical and radiologic anatomy : SRA*, 37(8), 1-3. doi:10.1007/s00276-014-1408-1
5. **Lee, H.**, Park, J.-H., Seo, I., Park, S.-H., Kim, S. (2014). Improved application of the electrophoretic tissue clearing technology, CLARITY, to intact solid organs including brain, pancreas, liver, kidney, lung, and intestine. *BMC developmental biology*, 14(1), 48. doi:10.1186/s12861-014-0048-3
4. Park, S.-H., Ryu, S.-Y., Yu, W.-J., Han, Y. E., Ji, Y.-S., Oh, K., Sohn, J.-W., Lim, A., Jeon, J.-P., **Lee, H.**, Lee, K.-H., Lee, S.-H., Berggren, P.-O., Jeon, J.-H., Ho, W.-K. (2013). Leptin promotes K(ATP) channel trafficking by AMPK signaling in pancreatic  $\beta$ -cells.. *Proceedings of the National Academy of Sciences of the United States of America*, 110(31), 12673-12678. doi.org/10.1073/pnas.1216351110
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1. He, X.-B., Yi, S.-H., Rhee, Y.-H., Kim, H., Han, Y.-M., Lee, S.-H., **Lee, H.**, Park, C.-H., Lee, Y.-S., Richardson, E., Kim, B.-W., Lee, S.-H. (2011). Prolonged membrane depolarization enhances midbrain dopamine neuron differentiation via epigenetic histone modifications.. *Stem cells (Dayton, Ohio)*, 29(11), 1861-1873.